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Abstract

The Family Expense Prediction System is a machine learning-based application designed to help families track, analyze, and predict their monthly expenses efficiently. Financial management is an essential aspect of maintaining household stability, yet many families struggle to monitor their expenses and predict future spending. Traditional methods such as manual record keeping are inefficient and do not provide meaningful insights into financial patterns. This project addresses this problem by implementing an automated system using machine learning techniques to predict total family expenses based on various expense categories.

The system uses a dataset containing expense records such as food expenses, rent expenses, travel expenses, shopping expenses, and bill expenses. The collected data is processed using Python and stored in MongoDB, a NoSQL database that ensures efficient data storage and retrieval. Data preprocessing techniques are applied to prepare the data for machine learning analysis.

Linear Regression, a supervised machine learning algorithm, is used to predict the total expense based on individual expense categories. The system analyzes the relationship between input variables and the total expense to generate accurate predictions. A frontend dashboard is developed using HTML and CSS, which allows users to input expense values and view predicted results.

Graphical visualization techniques such as bar charts are used to represent expense distribution and help users understand their spending patterns. The results demonstrate that machine learning can be effectively applied to predict family expenses and assist in financial planning. This system provides a practical solution for expense prediction and financial analysis and can be further enhanced for real-world applications.

Introduction

Financial management plays a vital role in maintaining the stability and well-being of a family. Every family spends money on various daily and monthly needs such as food, rent, travel, shopping, and utility bills. Proper tracking and analysis of these expenses are important to ensure effective budgeting and financial planning. However, many families still use traditional methods such as writing expenses in notebooks or using basic spreadsheets. These methods are inefficient, time-consuming, and do not provide accurate analysis or prediction of future expenses.

With the advancement of technology, Machine Learning has become an important tool for analyzing data and predicting future outcomes. Machine Learning is a branch of Artificial Intelligence that enables computer systems to learn from historical data and make predictions without explicit programming. By analyzing past expense data, machine learning algorithms can identify patterns and relationships between different expense categories and predict the total expense. This helps users understand their spending behavior and make better financial decisions.

The Family Expense Prediction System is developed to provide an efficient and intelligent solution for tracking and predicting family expenses. The system collects expense data including food expenses, rent expenses, travel expenses, shopping expenses, and bills. The collected data is processed using Python programming language and stored in MongoDB database, which provides efficient data storage, retrieval, and management. MongoDB is a NoSQL database that allows flexible storage of expense records and supports real-time data access.

The system uses Linear Regression, which is a supervised machine learning algorithm used for predicting continuous numerical values. Linear Regression analyzes the relationship between independent variables such as different expense categories and the dependent variable, which is the total expense. Based on the input provided by the user, the system predicts the total family expense accurately.

A user-friendly dashboard is developed using HTML and CSS, where users can enter expense details and view the predicted total expense. The system also provides graphical visualization in the form of bar charts, which helps users understand the distribution of expenses across different categories. This graphical analysis makes it easier to identify major expense areas and improve financial planning.

This project demonstrates the practical implementation of machine learning, database integration, and frontend development to solve real-world financial problems. The Family Expense Prediction System helps users efficiently track their expenses, analyze spending patterns, and predict future expenses, thereby improving overall financial management.

Problem Statement

Managing household expenses has become increasingly challenging due to rising living costs, irregular income patterns, and unplanned expenditures. Many families rely on manual recordkeeping or basic spreadsheet tools, which do not provide intelligent insights, future expense predictions, or financial planning support. As a result, families often struggle with budgeting, savings planning, and controlling unnecessary spending.

The Family Expense Predictor System aims to develop a smart, data-driven solution that analyzes historical household expense data and predicts future expenditures using machine learning techniques. The system will help families track income and expenses, categorize spending patterns, forecast upcoming monthly expenses, and provide insights for better financial decision-making.

By leveraging data analytics and predictive modeling, this system will:

- Identify spending trends and patterns
- Predict future monthly expenses based on past data
- Assist in budget planning and savings management
- Provide visual reports and dashboards for financial clarity

The proposed solution addresses the limitations of traditional expense tracking methods by integrating automation, predictive analytics, and user-friendly interfaces to enhance financial stability and planning for families.

Dataset Description – Family Expense Predictor System

The dataset used for the **Family Expense Predictor System** consists of historical household financial records collected over a specific period (monthly or daily basis). The dataset contains structured information about income, expenses, categories of spending, and savings patterns. This data is used to analyze spending behaviour and train machine learning models for predicting future expenses.

1. Data Source

• The expense data was first created and stored in a CSV file containing different family expense categories.

• The CSV file was converted into JSON format using Python for better data handling.

• The JSON data was inserted into MongoDB database, which is used to store and retrieve the dataset for prediction and analysis.

2. Dataset Features (Attributes)

The dataset contains the following features:

- Food Expense – Amount spent on food
- Rent Expense – Amount spent on house rent
- Travel Expense – Amount spent on transportation
- Shopping Expense – Amount spent on shopping
- Bills Expense – Amount spent on electricity, water, and other bills
- Total Expense – Total sum of all expenses (Target variable)

These features are used as input and output variables for training the machine learning model.

3. Data Characteristics

- The dataset is structured and numerical in nature.
- Each record represents family expenses for a specific period.
- The dataset contains multiple expense categories as independent variables.
- The Total Expense is the dependent variable used for prediction.
- The dataset is suitable for regression analysis using Linear Regression algorithm.

4. Data Preprocessing Steps

Before training the model, the dataset was preprocessed using Python. The preprocessing steps include:

- Loading the dataset from CSV file
- Converting CSV data into JSON format
- Inserting the dataset into MongoDB database
- Extracting the required features from the dataset
- Separating input variables and target variable
- Preparing the dataset for training the Linear Regression model

These steps ensure the dataset is clean and suitable for machine learning.

5. Dataset Size

- The dataset contains multiple records of family expenses with different expense values. Each record includes expense values for food, rent, travel, shopping, and bills along with total expense.
- The dataset size is sufficient to train and test the Linear Regression model for expense prediction.

6. Purpose of Dataset

The main purpose of the dataset is to train a machine learning model to predict the total family expense based on different expense categories.

The dataset helps in:

- Training the Linear Regression model
- Predicting total expenses based on input values
- Analyzing family spending patterns
- Helping users manage and plan their expenses

Thus, the dataset plays an important role in building an accurate Family Expense Prediction System.

Algorithms Used : Linear Regression

Linear Regression is a supervised machine learning algorithm used for predicting continuous numerical values based on the relationship between independent variables and a dependent variable. It is one of the simplest and most commonly used algorithms for prediction and forecasting tasks. Linear Regression works by identifying a linear relationship between input features and the output value, allowing the system to make accurate predictions based on historical data.

In the Family Expense Prediction System, Linear Regression is used to predict the total family expense based on different expense categories such as food expenses, rent expenses, travel expenses, shopping expenses, and bills. These expense categories act as independent variables

(input features), and the total expense acts as the dependent variable (output). The algorithm analyses the relationship between these variables and learns from the existing dataset to predict the total expense when new input values are provided.

Linear Regression works based on a mathematical equation that represents the relationship between variables. The equation of Linear Regression is:

$Y = a + bX$

Where:

Y = Predicted value (Total Expense)

X = Independent variable (Expense categories such as food, rent, travel, shopping, bills)

a = Intercept (constant value)

b = Slope (coefficient that determines the relationship between X and Y)

The algorithm calculates the best-fit line that minimizes the error between the actual values and the predicted values. This process is known as minimizing the error or loss function. Once the model is trained using the dataset, it can predict the total expense accurately based on new input values provided by the user.

In this project, the Linear Regression model is implemented using Python programming language. The expense dataset is first collected and preprocessed to ensure it is in the correct format. The dataset includes multiple expense categories and their corresponding total expense values. The Linear Regression algorithm is applied to this dataset to train the prediction model. After training, the model is used to predict the total family expense based on user input provided through the dashboard.

Implementation

Index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Family Expense Tracking System</title>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<style>
body{
    font-family: Arial;
    background: linear-gradient(to right, #4facfe, #00f2fe);
    margin:0;
}

.header{
    text-align:center;
    color:white;
    padding:20px;
}

.container{
    width:80%;
    margin:auto;
}

.card{
    background:white;
    padding:20px;
    margin-top:20px;
    border-radius:10px;
    box-shadow:0 0 10px rgba(0,0,0,0.2);
}

input,button{
    width:100%;
    padding:10px;
    margin-top:10px;
}

button{
    background:#007bff;
    color:white;
    border:none;
}

.result{
    font-size:20px;
    color:green;
    font-weight:bold;
    text-align:center;
}
</style>

</head>
```

```html
<html lang="en">
<body>
<div class="container">
<div class="card">

<h3>Expense Bar Chart</h3>

<canvas id="barChart"></canvas>

</div>

</div>

<script>

var expenseData = JSON.parse('{{ [food|default(0), rent|default(0), travel|default(0), shopping|default(0), bills|default(0)] | tojson | safe }}');

var labels = ["Food","Rent","Travel","Shopping","Bills"];

var ctx = document.getElementById("barChart").getContext("2d");

new Chart(ctx,{
    type:"bar",
    data:{
        labels:labels,
        datasets:[{
            label:"Expense Amount (₹)",
            data:expenseData,
            backgroundColor:[
                "#ff6384",
                "#36a2eb",
                "#4bc0c0",
                "#ff9f40",
                "#9966ff"
            ]
        }]
    },
    options:{
        responsive:true,
        scales:{
            y:{
                beginAtZero:true
            }
        }
    }
});

</script>

</body>
</html>
```

```html
<html lang="en">
<body>

<div class="header">
<h1>Family Expense Tracking System Dashboard</h1>
</div>

<div class="container">

<div class="card">

<form action="/predict" method="post">

<label>Month:</label>
<input type="number" name="month" required>

<label>Food Expense:</label>
<input type="number" name="food" required>

<label>Rent Expense:</label>
<input type="number" name="rent" required>

<label>Travel Expense:</label>
<input type="number" name="travel" required>

<label>Shopping Expense:</label>
<input type="number" name="shopping" required>

<label>Bills Expense:</label>
<input type="number" name="bills" required>

<button type="submit">Predict Total Expense</button>

</form>

</div>

<div class="card result">
{{ prediction_text }}
</div>

<div class="card">

<h3>Expense Bar Chart</h3>

<canvas id="barChart"></canvas>

</div>

</div>

<script>

var expenseData = JSON.parse('{{ [food|default(0), rent|default(0), travel|default(0), shopping|default(0), bills|default(0)] | tojson | safe }}');
```

expenses.csv

	A	B	C	D	E	F	G
1	month	food	rent	travel	shopping	bills	total_expense
2	1	5000	10000	2000	3000	2500	22500
3	2	5200	10000	1800	3500	2600	23100
4	3	4800	10000	2200	2800	2400	22200
5	4	5500	10000	2500	3200	2700	23900
6	5	5300	10000	2100	3100	2600	23100
7	6	6000	10000	2700	3500	2800	25000
8	7	5800	10000	2400	3300	2700	24200
9	8	6100	10000	2900	3700	2900	25600
10	9	5900	10000	2600	3400	2800	24700
11	10	6200	10000	3000	3800	3000	26000
12							

mongodb_insert.py

mongodb_insert.py

```python

# Create database
db = client["family_expense_db"]

# Create collection
collection = db["expenses"]

# Load JSON file
with open("expenses.json") as file:
    data = json.load(file)

# Delete old data
collection.delete_many({})

# Insert new data
collection.insert_many(data)

print("Data inserted into MongoDB successfully!")
```

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE | 25 | 1 - 10 of 10

```
_id: ObjectId('69a10b945547a2f90e5d678b')
month : 1
food : 5000
rent : 10000
travel : 2000
shopping : 3000
bills : 2500
total_expense : 22500

_id: ObjectId('69a10b945547a2f90e5d678c')
month : 2
food : 5200
rent : 10000
travel : 1800
shopping : 3500
bills : 2600
total_expense : 23100

_id: ObjectId('69a10b945547a2f90e5d678d')
month : 3
food : 4800
rent : 10000
travel : 2200
shopping : 2800
bills : 2400
total_expense : 22200
```

Ml_model.py

ml_model.py

```python
from flask import Flask, render_template, request
from pymongo import MongoClient
import pandas as pd
from sklearn.linear_model import LinearRegression

app = Flask(__name__)

# Connect to MongoDB
client = MongoClient("mongodb://localhost:27017/")
db = client["family_expense_db"]
collection = db["expenses"]

data = list(collection.find())
df = pd.DataFrame(data)

if "_id" in df.columns:
    df.drop("_id", axis=1, inplace=True)

X = df[["month", "food", "rent", "travel", "shopping", "bills"]]
y = df["total_expense"]

model = LinearRegression()
model.fit(X, y)

@app.route("/")
def home():
    return render_template("index.html")

@app.route("/predict", methods=["POST"])
def predict():
    month = float(request.form["month"])
    food = float(request.form["food"])
    rent = float(request.form["rent"])
    travel = float(request.form["travel"])
    shopping = float(request.form["shopping"])
    bills = float(request.form["bills"])

    user_input = [[month, food, rent, travel, shopping, bills]]
    prediction = model.predict(user_input)

    return render_template("index.html",
                           prediction_text=f"Predicted Expense: {round(prediction[0],2)}")

if __name__ == "__main__":
    app.run(debug=True)
```

Train_model_mongodb.py

train_model_mongodb.py

```python
from pymongo import MongoClient
import pandas as pd
from sklearn.linear_model import LinearRegression
import pickle

# Connect to MongoDB
client = MongoClient("mongodb://localhost:27017/")
db = client["family_expense_db"]
collection = db["expenses"]

# Get data
data = list(collection.find())

# Convert to DataFrame
df = pd.DataFrame(data)

# Remove _id column
if "_id" in df.columns:
    df.drop("_id", axis=1, inplace=True)

# Use only expense columns (NOT month)
X = df[['food', 'rent', 'travel', 'shopping', 'bills']]
y = df['total_expense']

# Train model
model = LinearRegression()
model.fit(X, y)

# Save model
pickle.dump(model, open("model.pkl", "wb"))

print("Model retrained successfully!")
```

Convert_csv_to_json.py

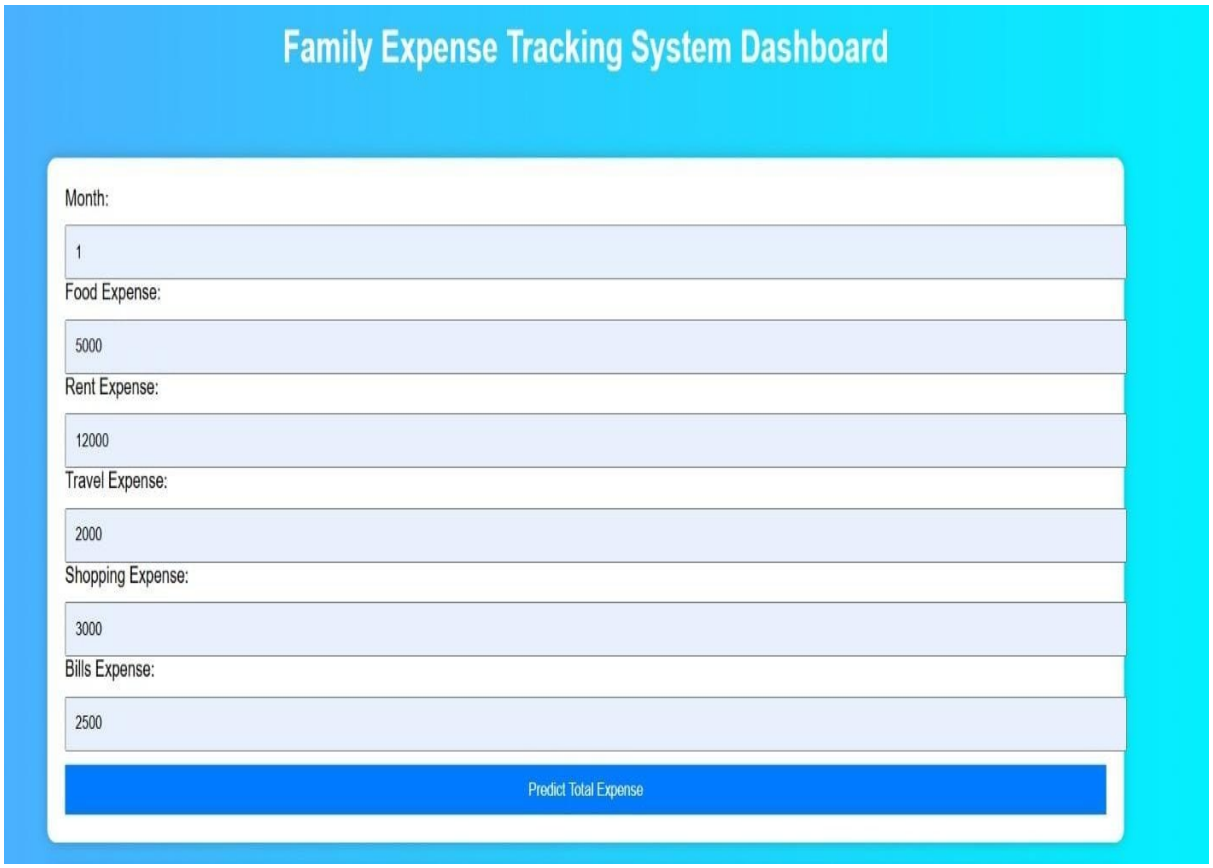
```python
convert_csv_to_json.py
import pandas as pd

# Load CSV file
df = pd.read_csv("expenses.csv")

# Remove missing values
df = df.dropna()

# Convert to JSON
df.to_json("expenses.json", orient="records", indent=4)

print("CSV converted to JSON successfully!")
```

app.py Train_model_mongodb.py

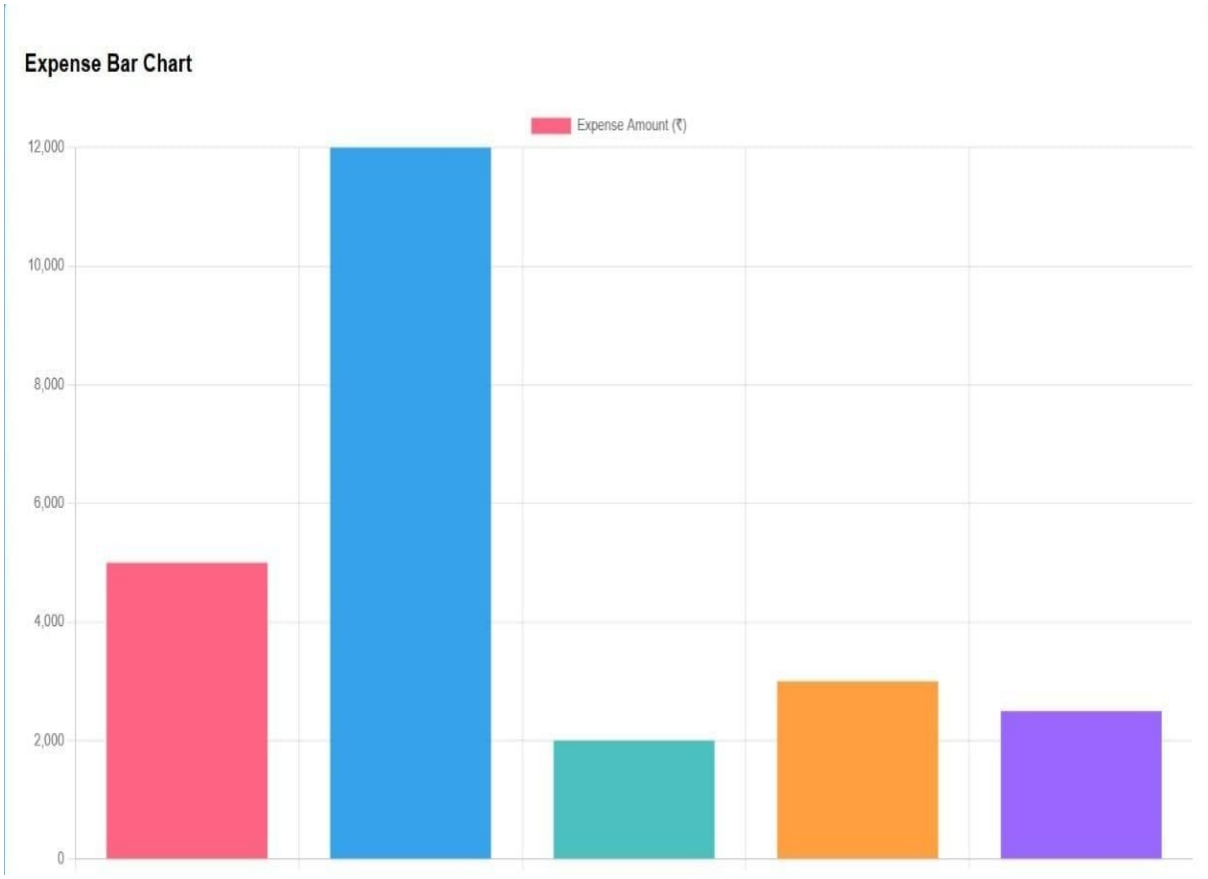
```python
app.py
from flask import Flask, render_template, request
import pickle
import numpy as np

# create flask app
app = Flask(__name__)

# load trained ML model
model = pickle.load(open('model.pkl', 'rb'))

# home page
@app.route('/')
def home():
    return render_template('index.html')

# prediction function
@app.route('/predict', methods=['POST'])
def predict():

    # get values from form
    month = request.form['month']
    food = float(request.form['food'])
    rent = float(request.form['rent'])
    travel = float(request.form['travel'])
    shopping = float(request.form['shopping'])
    bills = float(request.form['bills'])

    # prediction using model
    prediction = model.predict([[food, rent, travel, shopping, bills]])

    output = round(prediction[0], 2)
```

```python
app.py
def predict():

    # prediction using model
    prediction = model.predict([[food, rent, travel, shopping, bills]])

    output = round(prediction[0], 2)

    # send values back to html for graphs
    return render_template('index.html',
                           prediction_text="Predicted Total Expense: ₹" + str(output),
                           month=month,
                           food=food,
                           rent=rent,
                           travel=travel,
                           shopping=shopping,
                           bills=bills)

# run app
if __name__ == "__main__":
    app.run(debug=True)
```

Output:



Graphical Analysis



Graphical analysis helps visualize family spending patterns and understand trends.

•A Bar Chart is used to visually represent the family expenses across different categories such as Food, Rent, Travel, Shopping, and Bills.

• Each bar represents a specific expense category, and the height of the bar shows the amount spent in that category.

• The Bar Chart helps users compare expenses easily and identify which category has the highest and lowest spending.

• This graphical representation helps users understand their spending pattern and manage their expenses more effectively.

These visuals make it easier to track spending, plan budgets, and forecast future expenses effectively.

Conclusion

The Family Expense Predictor System provides an effective solution for managing household finances by analyzing historical income and expense data. It helps families track spending patterns, identify high-cost categories, and understand overall financial behavior.

This System successfully demonstrates the application of data analysis and machine learning in personal financial management. By collecting and analyzing historical income and expense data, the system can identify trends, high-spending categories, and irregular expenditures. The predictive models, such as Linear Regression and Random Forest, provide accurate forecasts of future monthly expenses, allowing families to plan budgets more effectively and make informed financial decisions.

Graphical analysis further enhances understanding by visually representing income, expenses, and savings patterns, making it easier to track financial health. Overall, the system not only aids in monitoring and controlling daily expenses but also promotes better financial discipline, reduces unnecessary spending, and helps families achieve long-term savings goals. It highlights how intelligent, data-driven tools can simplify household finance management and empower users to take proactive control of their finances.

Using predictive models like Linear Regression and Random Forest, the system forecasts future monthly expenses, enabling better budget planning and proactive financial decisionmaking. These predictions help families prepare for upcoming expenses and avoid overspending.

Overall, the system simplifies expense management, promotes better financial discipline, and supports long-term savings. It demonstrates how data-driven tools can empower families to plan, save, and make informed financial decisions efficiently.

References

1. Brownlee, J. (2019). *Machine Learning Mastery with Python: Understand Your Data, Create Accurate Models, and Work Projects End-to-End*. Machine Learning Mastery.
2. Aggarwal, C. C. (2018). *Data Mining: The Textbook*. Springer.
3. Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques* (3rd Edition). Morgan Kaufmann.
4. Geron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*. O'Reilly Media.
5. Kaur, P., & Arora, A. (2020). "A Review on Personal Expense Prediction Using Machine Learning Techniques." *International Journal of Computer Applications*, 176(2), 1–5.
6. Pandey, R., & Sharma, P. (2021). "Predicting Household Expenditure Using Data Mining Techniques." *Journal of Statistics and Management Systems*, 24(3), 271–281.
7. Raschka, S., & Mirjalili, V. (2019). *Python Machine Learning* (3rd Edition). Packt Publishing.
8. Online Resources:
   - https://www.kaggle.com/datasets (for sample financial/expense datasets) ○ https://scikit-learn.org/stable/ (for machine learning algorithms) ○ https://matplotlib.org/stable/gallery/index.html (for graphical visualization)